

# \* Ada Boost Machine Learning Algorithm:

M-33

Decision Trees

{Base learners}

Ensemble Techniques

{Weak learners}

Bagging

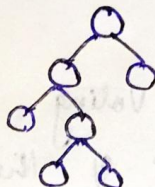
- I. Random Forest Classifier
- II. Random Forest Regressor

Decision Tree:

Overfitting:-

Train Acc ↑  
Test Acc ↓

low Bias  
High Variance



Boosting

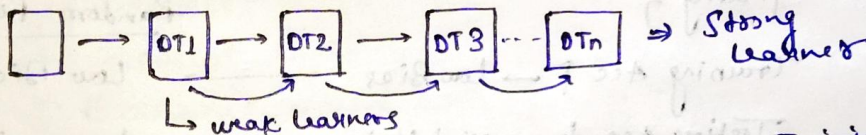
- I. Ada Boost
- II. Gradient Boosting
- III. XG Boost

Random Forest

{Bagging}

low Bias  
low Variance

\* Boosting {Sequentially connected}



weak learners: → haven't learn much from the Training Dataset

Random forest: → majority voting classifier Average of (Op)

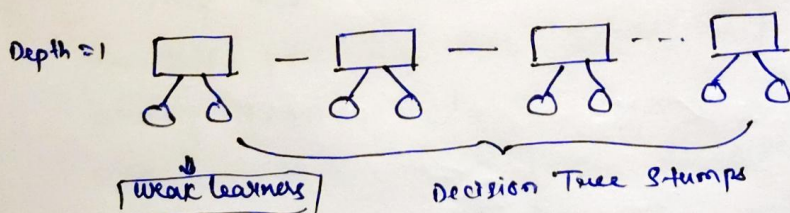
Ada Boost: → Assignment weights to the weak learners.

$M_1, \dots, M_n \rightarrow$  Decision Tree Stumps

$$F = \alpha_1(M_1) + \alpha_2(M_2) + \alpha_3(M_3) + \dots + \alpha_n(M_n)$$

$\{\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n\} \Rightarrow$  weights

Decision Stumps → means depth is 1



↳ Underfitting

↳ Train Acc ↓ 40%

↳ Test Acc ↓ 45%

High Bias

Low Variance

$\Rightarrow$  { low Bias  
high Variance }